

1 Composite Numerical Integration

The Newton-Cotes formulas are generally unsuitable for use over large integration intervals. High-degree formulas would be required, and the values of the coefficients in these formulas are difficult to obtain. Also, the Newton-Cotes formulas are based on interpolatory polynomials that use equally-spaced nodes, a procedure that is inaccurate over large intervals because of the oscillatory nature of high-degree polynomials.

In this section, we discuss a piecewise approach to numerical integration that uses the low-order Newton-Cotes formulas. These are the techniques most often applied.

Example 1. Use Simpson's rule to approximate $\int_0^4 e^x dx$ and compare this to the results obtained by adding the Simpson's rule approximations for $\int_0^2 e^x dx$ and $\int_2^4 e^x dx$. Compare these approximations to the sum of Simpson's rule for $\int_0^1 e^x dx$, $\int_1^2 e^x dx$, $\int_2^3 e^x dx$, and $\int_3^4 e^x dx$.

Solution: Simpson's rule on $[0, 4]$ uses $h = 2$ and gives

$$\int_0^4 e^x dx \approx \frac{2}{3} (e^0 + 4e^2 + e^4) = 56.76958$$

The exact answer in this case is $e^4 - e^0 = 53.59815$, and the error -3.17143 is far larger than we would normally accept.

Applying Simpson's rule on each of the intervals $[0, 2]$ and $[2, 4]$ uses $h = 1$ and gives

$$\begin{aligned} \int_0^4 e^x dx &= \int_0^2 e^x dx + \int_2^4 e^x dx \\ &\approx \frac{1}{3} (e^0 + 4e + e^2) + \frac{1}{3} (e^2 + 4e^3 + e^4) \\ &= \frac{1}{3} (e^0 + 4e + 2e^2 + 4e^3 + e^4) \\ &= 53.86385 \end{aligned}$$

The error has been reduced to -0.26570 .

For the integrals on $[0, 1]$, $[1, 2]$, $[2, 3]$, and $[3, 4]$ we use Simpson's rule four times with $h = \frac{1}{2}$ giving

$$\begin{aligned}
\int_0^4 e^x dx &= \int_0^1 e^x dx + \int_1^2 e^x dx + \int_2^3 e^x dx + \int_3^4 e^x dx \\
&\approx \frac{1}{6} (e_0 + 4e^{1/2} + e) + \frac{1}{6} (e + 4e^{3/2} + e^2) \\
&\quad + \frac{1}{6} (e^2 + 4e^{5/2} + e^3) + \frac{1}{6} (e^3 + 4e^{7/2} + e^4) \\
&= \frac{1}{6} (e^0 + 4e^{1/2} + 2e + 4e^{3/2} + 2e^2 + 4e^{5/2} + 2e^3 + 4e^{7/2} + e^4) \\
&= 53.61622
\end{aligned}$$

The error for this approximation has been reduced to -0.01807 .

To generalize this procedure for an arbitrary integral $\int_a^b f(x)dx$, choose an even integer n . Subdivide the interval $[a, b]$ into n subintervals, and apply Simpson's rule on each consecutive pair of subintervals. (See Figure 1)

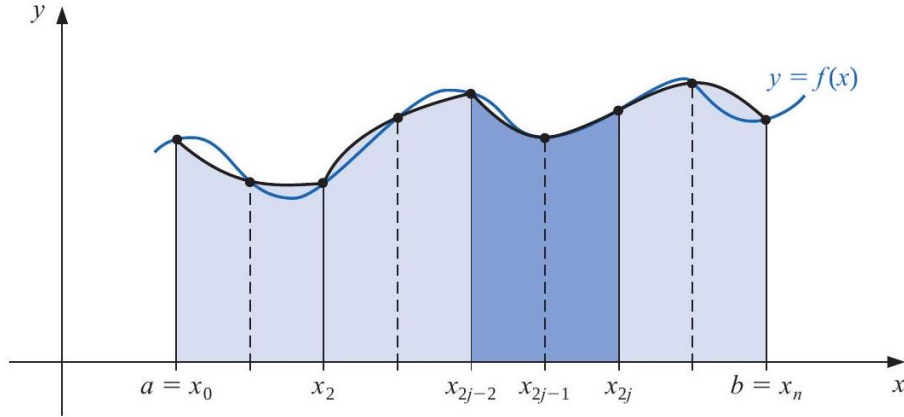


Figure 1: Figure 1

With $h = (b - a)/n$ and $x_j = a + jh$, for each $j = 0, 1, \dots, n$, we have

$$\begin{aligned}
\int_a^b f(x)dx &= \sum_{j=1}^{n/2} \int_{x_{2j-2}}^{x_{2j}} f(x)dx \\
&= \sum_{j=1}^{n/2} \left\{ \frac{h}{3} [f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})] - \frac{h^5}{90} f^{(4)}(\xi_j) \right\}
\end{aligned}$$

for some ξ_j with $x_{2j-2} < \xi_j < x_{2j}$, provided that $f \in C^4[a, b]$. Using the fact that for each $j = 1, 2, \dots, (n/2) - 1$ we have $f(x_{2j})$ appearing in the term

corresponding to the interval $[x_{2j-2}, x_{2j}]$ and also in the term corresponding to the interval $[x_{2j}, x_{2j+2}]$, we can reduce this sum to

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(x_0) + 2 \sum_{j=1}^{(n/2)-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(x_n) \right] - \frac{h^5}{90} \sum_{j=1}^{n/2} f^{(4)}(\xi_j)$$

The error associated with this approximation is

$$E(f) = -\frac{h^5}{90} \sum_{j=1}^{n/2} f^{(4)}(\xi_j)$$

where $x_{2j-2} < \xi_j < x_{2j}$, for each $j = 1, 2, \dots, n/2$.

These observations produce the following result.

Theorem

Let $f \in C^4[a, b]$, n be even, $h = (b - a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists a $\mu \in (a, b)$ for which the Composite Simpson's rule for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(a) + 2 \sum_{j=1}^{(n/2)-1} f(x_{2j}) + 4 \sum_{j=1}^{n/2} f(x_{2j-1}) + f(b) \right] - \frac{b-a}{180} h^4 f^{(4)}(\mu)$$

Notice that the error term for the Composite Simpson's rule is $O(h^4)$, whereas it was $O(h^5)$ for the standard Simpson's rule. However, these rates are not comparable because for standard Simpson's rule we have h fixed at $h = (b - a)/2$, but for Composite Simpson's rule we have $h = (b - a)/n$, for n an even integer. This permits us to considerably reduce the value of h when the Composite Simpson's rule is used.

The subdivision approach can be applied to any of the Newton-Cotes formulas. The extensions of the Trapezoidal (see Figure 4.8) and Midpoint rules are given without proof. The Trapezoidal rule requires only one interval for each application, so the integer n can be either odd or even.

Theorem

Let $f \in C^2[a, b]$, $h = (b-a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists a $\mu \in (a, b)$ for which the **Composite Trapezoidal rule** for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{2} \left[f(a) + 2 \sum_{j=1}^{n-1} f(x_j) + f(b) \right] - \frac{b-a}{12} h^2 f''(\mu)$$

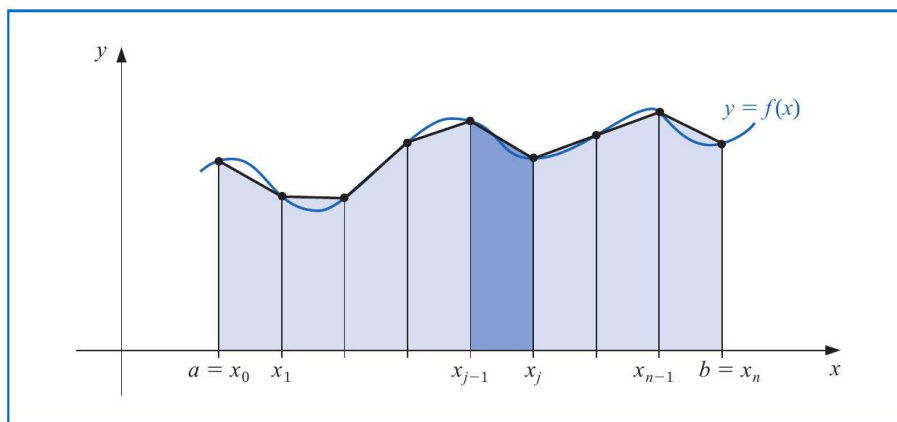


Figure 2: Figure 2

For the Composite Midpoint rule, n must again be even. (See Figure 3.)

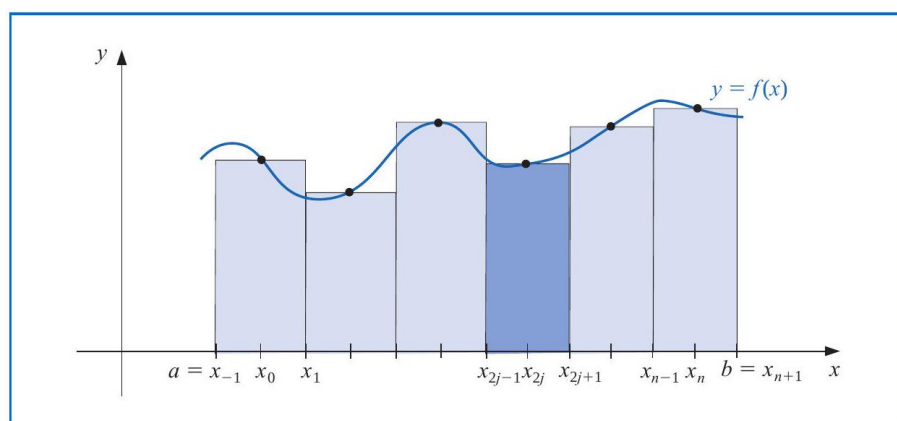


Figure 3: Figure 3

Theorem

Let $f \in C^2[a, b]$, n be even, $h = (b - a)/(n + 2)$, and $x_j = a + (j + 1)h$ for each $j = -1, 0, \dots, n + 1$. There exists a $\mu \in (a, b)$ for which the Composite Midpoint rule for $n + 2$ subintervals can be written with its error term as

$$\int_a^b f(x)dx = 2h \sum_{j=0}^{n/2} f(x_{2j}) + \frac{b-a}{6} h^2 f''(\mu)$$

Example 2. Determine values of h that will ensure an approximation error

of less than 0.00002 when approximating $\int_0^\pi \sin x dx$ and employing
(a) Composite Trapezoidal rule and (b) Composite Simpson's rule.

Solution:

(a) The error form for the Composite Trapezoidal rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^2}{12} f''(\mu) \right| = \left| \frac{\pi h^2}{12} (-\sin \mu) \right| = \frac{\pi h^2}{12} |\sin \mu|.$$

To ensure sufficient accuracy with this technique we need to have

$$\frac{\pi h^2}{12} |\sin \mu| \leq \frac{\pi h^2}{12} < 0.00002$$

Since $h = \pi/n$ implies that $n = \pi/h$, we need

$$\frac{\pi^3}{12n^2} < 0.00002 \quad \text{which implies that } n > \left(\frac{\pi^3}{12(0.00002)} \right)^{1/2} \approx 359.44$$

and the Composite Trapezoidal rule requires $n \geq 360$.

(b) The error form for the Composite Simpson's rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^4}{180} f^{(4)}(\mu) \right| = \left| \frac{\pi h^4}{180} \sin \mu \right| = \frac{\pi h^4}{180} |\sin \mu|.$$

To ensure sufficient accuracy with this technique we need to have

$$\frac{\pi h^4}{180} |\sin \mu| \leq \frac{\pi h^4}{180} < 0.00002$$

Using again the fact that $n = \pi/h$ gives

$$\frac{\pi^5}{180n^4} < 0.00002 \quad \text{which implies that } n > \left(\frac{\pi^5}{180(0.00002)} \right)^{1/4} \approx 17.07$$

So Composite Simpson's rule requires only $n \geq 18$.
Composite Simpson's rule with $n = 18$ gives

$$\int_0^\pi \sin x dx \approx \frac{\pi}{54} \left[2 \sum_{j=1}^8 \sin \left(\frac{j\pi}{9} \right) + 4 \sum_{j=1}^9 \sin \left(\frac{(2j-1)\pi}{18} \right) \right] = 2.0000104$$

This is accurate to within about 10^{-5} because the true value is $-\cos(\pi) - (-\cos(0)) = 2$.

Composite Simpson's rule is the clear choice if you wish to minimize computation. For comparison purposes, consider the Composite Trapezoidal rule using $h = \pi/18$ for the integral in Example 2. This approximation uses the same function evaluations as Composite Simpson's rule but the approximation in this case

$$\int_0^\pi \sin x dx \approx \frac{\pi}{36} \left[2 \sum_{j=1}^{17} \sin \left(\frac{j\pi}{18} \right) + \sin 0 + \sin \pi \right] = \frac{\pi}{36} \left[2 \sum_{j=1}^{17} \sin \left(\frac{j\pi}{18} \right) \right] = 1.9949205.$$

is accurate only to about 5×10^{-3} .

Round-Off Error Stability

In Example 2 we saw that ensuring an accuracy of 2×10^{-5} for approximating $\int_0^\pi \sin x dx$ required 360 subdivisions of $[0, \pi]$ for the Composite Trapezoidal rule and only 18 for Composite Simpson's rule. In addition to the fact that less computation is needed for the Simpson's technique, you might suspect that because of fewer computations this method would also involve less round-off error. However, an important property shared by all the composite integration techniques is a stability with respect to round-off error. That is, the round-off error does not depend on the number of calculations performed.

To demonstrate this rather amazing fact, suppose we apply the Composite Simpson's rule with n subintervals to a function f on $[a, b]$ and determine the maximum bound for the round-off error. Assume that $f(x_i)$ is approximated by $\tilde{f}(x_i)$ and that

$$f(x_i) = \tilde{f}(x_i) + e_i, \quad \text{for each } i = 0, 1, \dots, n,$$

where e_i denotes the round-off error associated with using $\tilde{f}(x_i)$ to approximate $f(x_i)$. Then the accumulated error, $e(h)$, in the Composite Simpson's rule is

$$\begin{aligned}
e(h) &= \left| \frac{h}{3} \left[e_0 + 2 \sum_{j=1}^{(n/2)-1} e_{2j} + 4 \sum_{j=1}^{n/2} e_{2j-1} + e_n \right] \right| \\
&\leq \frac{h}{3} \left[|e_0| + 2 \sum_{j=1}^{(n/2)-1} |e_{2j}| + 4 \sum_{j=1}^{n/2} |e_{2j-1}| + |e_n| \right].
\end{aligned}$$

If the round-off errors are uniformly bounded by ε , then

$$e(h) \leq \frac{h}{3} \left[\varepsilon + 2 \left(\frac{n}{2} - 1 \right) \varepsilon + 4 \left(\frac{n}{2} \right) \varepsilon + \varepsilon \right] = \frac{h}{3} 3n\varepsilon = nh\varepsilon.$$

But $nh = b - a$, so

$$e(h) \leq (b - a)\varepsilon$$

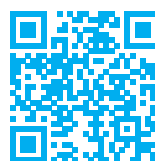
a bound independent of h (and n). This means that, even though we may need to divide an interval into more parts to ensure accuracy, the increased computation that is required does not increase the round-off error. This result implies that the procedure is stable as h approaches zero. Recall that this was not true of the numerical differentiation procedures we have considered.

2 Romberg Integration

This video presents a discussion on Romberg integration. Follow the discussion and then solve the problems in the subsequent quiz. **Video** Visit the

URL below to view a video:

<https://www.youtube.com/embed/oAh0qTJtSuk>



3 Adaptive Quadrature Methods

Read on adaptive quadrature method in the reference provided in the link below in the pages and then attempt the subsequent quiz.

[https://math.libretexts.org/Bookshelves/Calculus//CLP-2_Integral_Calculus_\(Feldman_Rech](https://math.libretexts.org/Bookshelves/Calculus//CLP-2_Integral_Calculus_(Feldman_Rech)

4 Gaussian Quadrature

The Newton-Cotes formulas in Section 4 of Module 4 were derived by integrating interpolating polynomials. The error term in the interpolating polynomial of degree n involves the $(n + 1)$ st derivative of the function being approximated, so a Newton-Cotes formula is exact when approximating the integral of any polynomial of degree less than or equal to n .

All the Newton-Cotes formulas use values of the function at equally-spaced points. This restriction is convenient when the formulas are combined to form the composite rules we considered in Section 1 of this Module, but it can significantly decrease the accuracy of the approximation. Consider, for example, the Trapezoidal rule applied to determine the integrals of the functions whose graphs are shown in Figure 4.

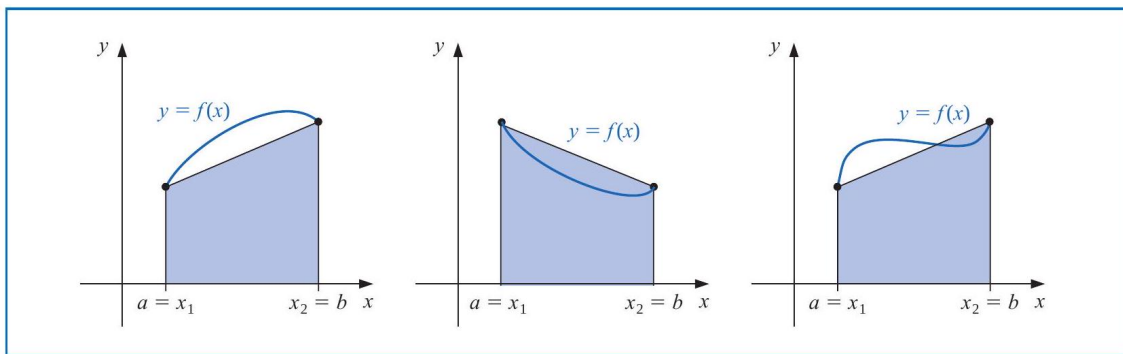


Figure 4: Figure 4

The Trapezoidal rule approximates the integral of the function by integrating the linear function that joins the endpoints of the graph of the function. But this is not likely the best line for approximating the integral. Lines such as those shown in Figure 5 would likely give much better approximations in most cases.

Gaussian quadrature chooses the points for evaluation in an optimal, rather than equally spaced, way. The nodes $x_1, x_2, \dots, x_n$ in the interval $[a, b]$ and coefficients $c_1, c_2, \dots, c_n$, are chosen to minimize the expected error obtained in the approximation

$$\int_a^b f(x) dx \approx \sum_{i=1}^n c_i f(x_i)$$

To measure this accuracy, we assume that the best choice of these values produces the exact result for the largest class of polynomials, that is, the choice that gives the greatest degree of precision.

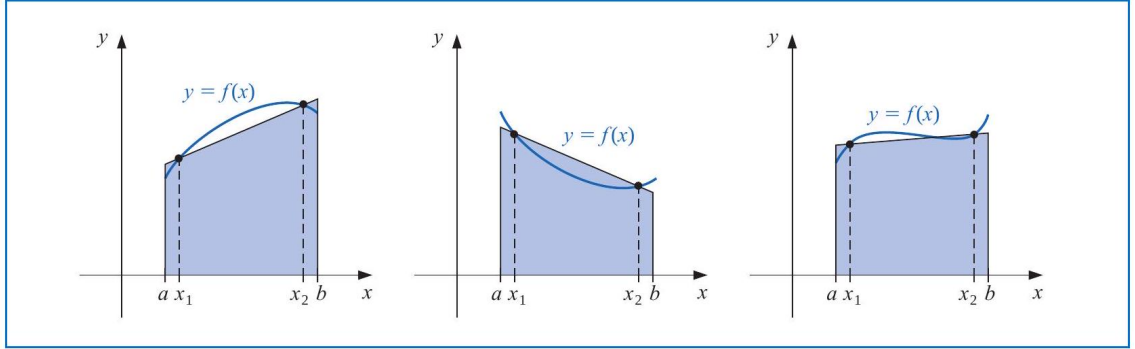


Figure 5: Figure 5

The coefficients $c_1, c_2, \dots, c_n$ in the approximation formula are arbitrary, and the nodes $x_1, x_2, \dots, x_n$ are restricted only by the fact that they must lie in $[a, b]$, the interval of integration. This gives us $2n$ parameters to choose. If the coefficients of a polynomial are considered parameters, the class of polynomials of degree at most $2n - 1$ also contains $2n$ parameters. This, then, is the largest class of polynomials for which it is reasonable to expect a formula to be exact. With the proper choice of the values and constants, exactness on this set can be obtained.

To illustrate the procedure for choosing the appropriate parameters, we will show how to select the coefficients and nodes when $n = 2$ and the interval of integration is $[-1, 1]$. We will then discuss the more general situation for an arbitrary choice of nodes and coefficients and show how the technique is modified when integrating over an arbitrary interval.

Suppose we want to determine c_1, c_2, x_1 , and x_2 so that the integration formula

$$\int_{-1}^1 f(x) dx \approx c_1 f(x_1) + c_2 f(x_2)$$

gives the exact result whenever $f(x)$ is a polynomial of degree $2(2) - 1 = 3$ or less, that is, when

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

for some collection of constants, a_0, a_1, a_2 , and a_3 . Because

$$\int (a_0 + a_1x + a_2x^2 + a_3x^3) dx = a_0 \int 1dx + a_1 \int xdx + a_2 \int x^2dx + a_3 \int x^3dx$$

this is equivalent to showing that the formula gives exact results when $f(x)$ is $1, x, x^2$, and x^3 . Hence, we need c_1, c_2, x_1 , and x_2 , so that

$$\begin{aligned} c_1 \cdot 1 + c_2 \cdot 1 &= \int_{-1}^1 1 dx = 2, & c_1 \cdot x_1 + c_2 \cdot x_2 &= \int_{-1}^1 x dx = 0 \\ c_1 \cdot x_1^2 + c_2 \cdot x_2^2 &= \int_{-1}^1 x^2 dx = \frac{2}{3}, & \text{and} \\ c_1 \cdot x_1^3 + c_2 \cdot x_2^3 &= \int_{-1}^1 x^3 dx = 0 \end{aligned}$$

A little algebra shows that this system of equations has the unique solution

$$c_1 = 1, \quad c_2 = 1, \quad x_1 = -\frac{\sqrt{3}}{3}, \quad \text{and} \quad x_2 = \frac{\sqrt{3}}{3},$$

which gives the approximation formula

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right) \quad (4.40)$$

This formula has degree of precision 3, that is, it produces the exact result for every polynomial of degree 3 or less.

Legendre Polynomials

The technique we have described could be used to determine the nodes and coefficients for formulas that give exact results for higher-degree polynomials, but an alternative method obtains them more easily. The set that is relevant to our problem is the Legendre polynomials, a collection $\{P_0(x), P_1(x), \dots, P_n(x), \dots\}$ with properties:

- (1) For each n , $P_n(x)$ is a monic polynomial of degree n .

Recall that monic polynomials have leading coefficient 1.

- (2) $\int_{-1}^1 P(x)P_n(x) dx = 0$ whenever $P(x)$ is a polynomial of degree less than n .

The first few Legendre polynomials are

$$\begin{aligned} P_0(x) &= 1, & P_1(x) &= x, & P_2(x) &= x^2 - \frac{1}{3}, \\ P_3(x) &= x^3 - \frac{3}{5}x, & \text{and} & & P_4(x) &= x^4 - \frac{6}{7}x^2 + \frac{3}{35}. \end{aligned}$$

The roots of these polynomials are distinct, lie in the interval $(-1, 1)$, have a symmetry with respect to the origin, and, most importantly, are the correct choice for determining the parameters that give us the nodes and coefficients for our quadrature method.

The nodes $x_1, x_2, \dots, x_n$ needed to produce an integral approximation formula that gives exact results for any polynomial of degree less than $2n$ are the roots of the n th-degree Legendre polynomial. This is established by the following result.

Theorem

Suppose that $x_1, x_2, \dots, x_n$ are the roots of the n th Legendre polynomial $P_n(x)$ and that for each $i = 1, 2, \dots, n$, the numbers c_i are defined by

$$c_i = \int_{-1}^1 \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx$$

If $P(x)$ is any polynomial of degree less than $2n$, then

$$\int_{-1}^1 P(x) dx = \sum_{i=1}^n c_i P(x_i)$$

The constants c_i needed for the quadrature rule can be generated from the equation in Theorem 4, but both these constants and the roots of the Legendre polynomials are extensively tabulated. Table 1 lists these values for $n = 2, 3, 4$, and 5 .

Example 3. Approximate $\int_{-1}^1 e^x \cos x dx$ using Gaussian quadrature with $n = 3$.

Solution: The entries in Table 1 give us

$$\begin{aligned} \int_{-1}^1 e^x \cos x dx &\approx 0.5e^{0.774596692} \cos 0.774596692 \\ &\quad + 0.8 \cos 0 + 0.5e^{-0.774596692} \cos(-0.774596692) \\ &= 1.9333904 \end{aligned}$$

Integration by parts can be used to show that the true value of the integral is 1.9334214 , so the absolute error is less than 3.2×10^{-5} .

n	Roots $r_{n,i}$	Coefficients $c_{n,i}$
2	0.5773502692	1.0000000000
2	-0.5773502692	1.0000000000
3	0.7745966692	0.5555555556
3	0.0000000000	0.8888888889
3	-0.7745966692	0.5555555556
4	0.8611363116	0.3478548451
4	0.3399810436	0.6521451549
4	-0.3399810436	0.6521451549
4	-0.8611363116	0.3478548451
5	0.9061798459	0.2369268850
5	0.5384693101	0.4786286705
5	0.0000000000	0.5688888889
5	-0.5384693101	0.4786286705
5	-0.9061798459	0.2369268850

Table 1: Table 1

Gaussian Quadrature on Arbitrary Intervals

An integral $\int_a^b f(x)dx$ over an arbitrary $[a, b]$ can be transformed into an integral over $[-1, 1]$ by using the change of variables (see Figure 6):

$$t = \frac{2x - a - b}{b - a} \iff x = \frac{1}{2}[(b - a)t + a + b]$$

This permits Gaussian quadrature to be applied to any interval $[a, b]$, because

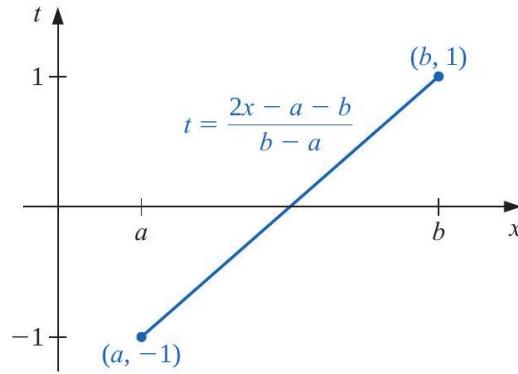


Figure 6: Figure 6

$$\int_a^b f(x)dx = \int_{-1}^1 f\left(\frac{(b-a)t + (b+a)}{2}\right) \frac{(b-a)}{2} dt \quad (1)$$

Example 4. Consider the integral $\int_1^3 x^6 - x^2 \sin(2x)dx = 317.3442466$.

- (a) Compare the results for the closed Newton-Cotes formula with $n = 1$, the open Newton-Cotes formula with $n = 1$, and Gaussian Quadrature when $n = 2$.
- (b) Compare the results for the closed Newton-Cotes formula with $n = 2$, the open Newton-Cotes formula with $n = 2$, and Gaussian Quadrature when $n = 3$.

Solution:

- (a) Each of the formulas in this part requires 2 evaluations of the function $f(x) = x^6 - x^2 \sin(2x)$. The Newton-Cotes approximations are

$$\text{Closed } n = 1 : \frac{2}{2}[f(1) + f(3)] = 731.6054420$$

$$\text{Open } n = 1 : \frac{3(2/3)}{2}[f(5/3) + f(7/3)] = 188.7856682$$

Gaussian quadrature applied to this problem requires that the integral first be transformed into a problem whose interval of integration is $[-1, 1]$. Using Eq.1 gives

$$\int_1^3 x^6 - x^2 \sin(2x)dx = \int_{-1}^1 (t+2)^6 - (t+2)^2 \sin(2(t+2))dt$$

Gaussian quadrature with $n = 2$ then gives

$$\int_1^3 x^6 - x^2 \sin(2x)dx \approx f(-0.5773502692 + 2) + f(0.5773502692 + 2) = 306.8199344;$$

- (b) Each of the formulas in this part requires 3 function evaluations. The Newton-Cotes approximations are

$$\text{Closed } n = 2 : \frac{(1)}{3}[f(1) + 4f(2) + f(3)] = 333.2380940$$

$$\text{Open } n = 2 : \frac{4(1/2)}{3}[2f(1.5) - f(2) + 2f(2.5)] = 303.5912023$$

Gaussian quadrature with $n = 3$, once the transformation has been done, gives

$$\begin{aligned} \int_1^3 x^6 - x^2 \sin(2x) dx \approx & 0.5 \bar{f}(-0.7745966692 + 2) + 0.8 \bar{f}(2) \\ & + 0.5 \bar{f}(0.7745966692 + 2) = 317.2641516 \end{aligned}$$

The Gaussian quadrature results are clearly superior in each instance.